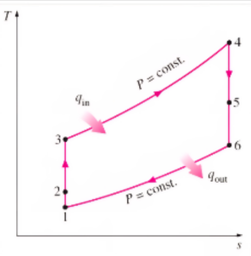
# Ideal jet




1-2: Isentropic compression in Diffuser
2-3: Isentropic compression in Compressor
3-4: Constant pressure heat addition
4-5: Isentropic expansion in Turbine
5-6: Isentropic expansion in Nozzle
6-1: Constant pressure heat rejection

$$\frac{T_1}{T_3} = \left(\frac{P_1}{P_3}\right)^{(k-1)/k}$$

Diffuser 1-2

$\cancel{q} - \cancel{w} = \dot{m}\left[(h_2 - h_1) + \frac{1}{2}(\cancel{V_2^2} - V_1^2)\right]$

Turbine 4-5

$w_{turb} = w_{comp}$

$C_p(T_5 - T_4) = C_p(T_3 - T_2)$

Nozzle 5-6

$\cancel{q} - \cancel{w} = \dot{m}\left[(h_6 - h_5) + \frac{1}{2}(V_6^2 - \cancel{V_5^2})\right]$

Thrust:

$F =$

Propulsive Power:

$\dot{W}_p = F\, v_{aircraft} = \dot{m}(v_{exit} - v_{inlet})\, v_{aircraft}$

Propulsive Efficiency:

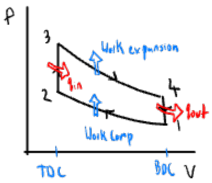
$\eta_p = \frac{\text{propulsive power}}{\text{heat input}} = \frac{\dot{W}_p}{\dot{Q}_{in}}$

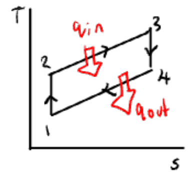
mass flowrate of fuel

$\dot{m} = \frac{\dot{Q}_{in}}{HV} \leftarrow$ heating value of fuel

# OTTO Cycle Petrol engine

4 stroke: 2 revolutions. 2 stroke: 1 revolution




$MEP = \frac{W_{net}}{V_1 - V_2} = \frac{W_{net}}{V_1(1 - \frac{1}{r})}$

$V_1 = \frac{RT_1}{P_1}$

CSH = use isentropic relationships, ideal gas equations, $c_p(\delta T)$

VSH = use tables find

1→2 Isentropic Compression

$\cancel{q_{12}} - W_{12} = u_2 - u_1 = C_v(T_2 - T_1)$ or $T_2 = T_1\left(\frac{V_1}{V_2}\right)^{k-1}$

2→3 Isochoric heating (constant volume)

$q_{23} - \cancel{W_{23}} = C_v(T_3 - T_2)$

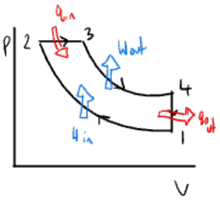
3→4 Isentropic expansion

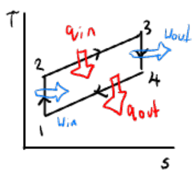
$-W_{34} = C_v(T_4 - T_3)$ or $\frac{T_4}{T_3} = \left(\frac{V_3}{V_4}\right)^{k-1}$

4→1 Isochoric heat rejection

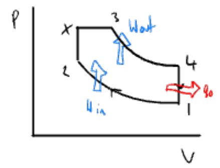
$q_{41} = q_{out} = C_v(T_1 - T_4)$

# Diesel Cycle




# Dual Diesel Cycle



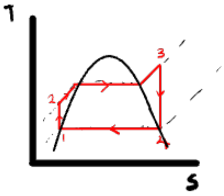

1→2 Isentropic Compression

2→3 Isobaric heat addition

3→4 Isentropic expansion (Power Stroke)

4→1 Constant Volume heat rejection (exhaust)

# Rankine Ideal Vapor Cycle - Steam power plant




1→2: Isentropic compression in the feedwater pump

2→3: Isobaric evaporation and super heating in the boiler

3→4: Isentropic expansion in the turbine

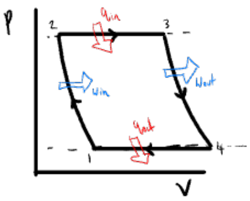
4→1: Isobaric condensation in the condenser

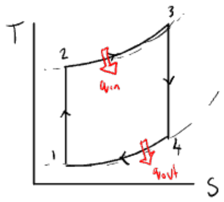
Boiler: $(w=0) \Rightarrow q_{in} = (h_3 - h_2)$

Turbine: $(q=0) \Rightarrow w_{turb} = (h_4 - h_3)$

Condenser: $(w=0) \Rightarrow q_{out} = (h_1 - h_4)$

# Brayton Cycle - Gas turbine engine



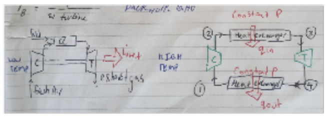




1-2 isentropic compression

2→3 isobaric heat addition

3→4 isentropic expansion

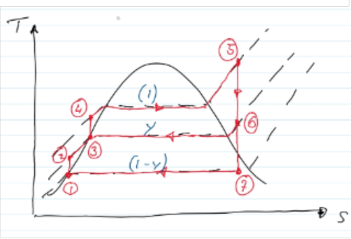
4→1 isobaric heat rejection



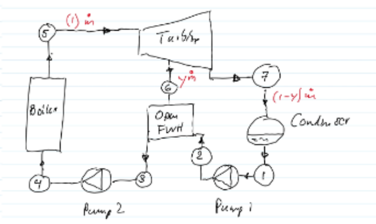
Backwork Ratio

$r_B = \frac{W_{comp}}{W_{turb}}$

# Ideal regenerative Rankine Cycle cycle




Open Feedwater Heater




State 1 SL

$x = 0$ $\quad V_1 =$

$P =$ $\quad h_1 =$

State 2

$W_{pump} = v_1(P_2 - P_1)\left(\frac{kJ}{kg}\right)$

$h_2 = h_1 + W_{pump}$

State 3 SL

$x = 0$ $\quad h_3 =$

$P =$ $\quad v_3 =$

State 4

$W_{pump2} = v_3(P_4 - P_3)$

$h_4 = W_{pump2} + h_3$

State 5 SH

$P =$ $\quad h_5 =$

$T =$ $\quad S_5 =$

State 6

$S_6 = S_5 <$ or $>$ than $Sg$ @ $P_6$

$S_6 < S_g$ = Sat Mixture A-5

$x = \frac{S_6 - S_f}{S_{fg}}$

$h_6 = h_f + x h_{fg}$

State 7

$P_7 =$ $\quad x_7 = \frac{S_7 - S_f}{S_{fg}}$

$S_7 = S_6 = S_5$ $\quad h_7 = h_f + x h_{fg}$


$y\cancel{\dot{m}}h_6 + (1-y)\cancel{\dot{m}}h_2 = \cancel{\dot{m}}h_3$

$y = \frac{h_3 - h_2}{h_6 - h_2}$ y = fraction of steam extracted by turbine

a) $W_{net} = q_{in} - q_{out}$

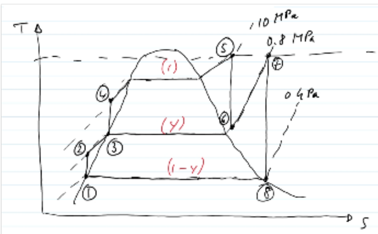
$q_{in} = h_5 - h_4$

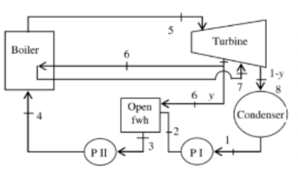
$q_{out} = (1-y)(h_7 - h_1)\left(\frac{kJ}{kg}\right)$

b) $\eta_{th} = \frac{W_{net}}{q_{in}}$

$P_2 = P_6$

Ideal reheat regenerative Rankine Cycle

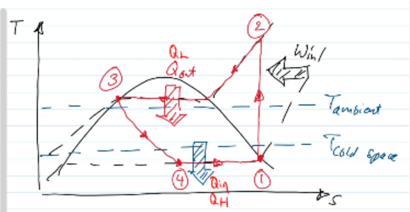



$q_{in} = (h_5 - h_4) + (1-y)(h_7 - h_6)$

State 8 $S_7 = S_8 > < S_g$ @ $P_8$

# Refrigeration




1→2: Isentropic compression in a compressor

2→3: Constant pressure heat rejection in the condenser

3→4: Isenthalpic (constant enthalpy) throttling

4→1: Isobaric heat absorption in the evaporator

$COP_R = \frac{Q_L}{W_{in}} = \frac{h_1 - h_4}{h_2 - h_1}$

$\Delta 10K \Rightarrow -10 \to -20°C$

$25°C \to 35°C$

State 1

$x = 1$ $\quad h_g =$

$S_g =$

State 2 Superheated

$S_2 = S_1$ $\quad h_2 =$

$P_2 =$

State 3

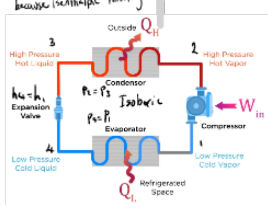
$x = 0$ $\quad h_3 =$

$P =$

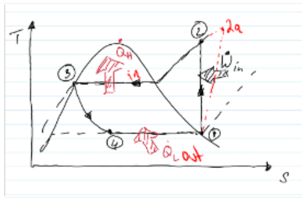
State 4

$h_3 = h_4$

because isenthalpic throttling




# Heatpump




$\eta_c = \frac{h_{2a} - h_1}{h_2 - h_1} = \frac{W_{net}}{q_{in}}$

## Interpolation

$$h_6 = h_1 + \frac{(h_2 - h_1)(s_a - s_1)}{(s_2 - s_1)}$$

## UNITS

| | | |
|---|---|---|
| Giga | G | $\times 10^9$ |
| mega | M | $\times 10^6$ |
| Kilo | K | $\times 10^3$ |
| milli | m | $\times 10^{-3}$ |
| micro | $\mu$ | $\times 10^{-6}$ |
| nano | n | $\times 10^{-9}$ |

0.8 MPa = 800 KPa

$$KPa \times \frac{m^3}{Kg} = \frac{KJ}{Kg}$$

$$\frac{KJ}{H} \div 3600 = \frac{KJ}{S}$$

$$\frac{KJ}{S} \div \frac{1}{3600} = \frac{KJ}{H}$$

Compressors

Isentropic Efficiency of Compressors

$$\eta_c = \frac{\text{isentropic work input}}{\text{actual work input}}$$

$$\boxed{\eta_c = \frac{h_{2s} - h_1}{h_{2a} - h_1}}$$

$P_2$

$$q - w = \dot{m}\left[(h_2 - h_1) + \frac{1}{2}(V_2^2 - V_1^2)\right]$$

Saturated water vapour $x = 1$

Turbines

$$\boxed{\eta_T = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}}$$ isentropic efficiency of turbines

Nozzles

$$\boxed{\eta_N = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}}$$ isentropic efficiency of a nozzle